

AKSHITH KALVAKOTA

Akshitkalvakota123@gmail.com

AI/ML Intern

Body Landmark Detection & Measurement Analysis using MediaPipe

The objective of this task was to evaluate the effectiveness of MediaPipe Pose for detecting human body landmarks and to analyze how extracted body measurements can contribute to improving the accuracy of AI-based virtual try-on systems. Multiple images with different poses were tested to observe landmark detection consistency, measurement reliability, and potential limitations.

Tools Used

- Python
- MediaPipe Pose
- OpenCV
- NumPy
- VS Code

Virtual try-on systems rely heavily on accurate body understanding to place garments realistically on a user's body. Before clothing can be aligned and fitted correctly, important body landmarks such as shoulders, hips, waist, elbows, knees, and ankles must be identified. MediaPipe Pose provides a lightweight solution for detecting these landmarks from a single RGB image.

This study focused on evaluating landmark detection performance across different poses and investigating how body measurements derived from landmarks can improve virtual garment fitting.

PROCESS:

Step 1: Image Collection

Several full-body images were collected with varying poses, including:

- Standing straight facing the camera
- Arms extended sideways
- Slight side pose
- Walking pose

The images were selected to represent common variations that may occur in real-world virtual try-on applications.

Step 2: Landmark Detection

MediaPipe Pose was applied to each image to detect 33 body landmarks. The detected landmarks were visualized and verified manually.

Step 3: Measurement Estimation

Using landmark coordinates, the following measurements were estimated:

- Shoulder Width
- Body Height
- Waist Position
- Hip Position
- Arm Length (Approximate)

Step 4: Accuracy Evaluation

Landmark visibility and positioning were analyzed across different poses to identify strengths and limitations of the detection model.

Body Landmarks Identified

MediaPipe successfully detected major body landmarks including:

- Nose
- Eyes
- Ears
- Shoulders
- Elbows
- Wrists
- Hips
- Knees
- Ankles
- Feet

The model consistently identified visible landmarks in front-facing images and maintained stable detection under moderate pose variations.

Shoulder Width Estimation

Shoulder width was calculated using the distance between the left and right shoulder landmarks.

We observed that

1. Front-facing images produced the most reliable shoulder width estimates.
2. Slight body rotations resulted in reduced apparent shoulder width due to perspective distortion.
3. Shoulder landmarks remained stable even when arms were raised.

Shoulder measurements showed good consistency and can be effectively used for garment scaling and alignment.

Height Estimation

Body height was estimated using the vertical distance between head-related landmarks and ankle landmarks.

We observed that

1. Accurate height estimation required the complete body to be visible in the image.
2. Cropped images significantly reduced measurement reliability.
3. Standing poses generated the best results.

Height estimation can provide useful scaling information for virtual try-on systems but may require calibration to convert pixel measurements into real-world units.

Waist Position Detection

The waist region was estimated using the relative positions of shoulder and hip landmarks.

We observed that

- Waist location remained stable in front-facing images.
- Side poses slightly affected waist center estimation.
- Loose clothing occasionally reduced landmark confidence around the torso.

Waist position estimation is sufficiently accurate for garment placement and body alignment tasks.

Pose Accuracy Observations

1. Standing Pose

- Highest landmark detection accuracy.
- Nearly all landmarks detected correctly.
- Best measurement consistency.

2. Arms Extended Pose

- Shoulder, elbow, and wrist landmarks detected accurately.
- Improved visibility of upper-body structure.

3. Side Pose

- Some landmarks partially occluded.
- Slight reduction in shoulder and waist estimation accuracy.

4. Walking Pose

- Dynamic limb positions occasionally reduced landmark stability.
- Lower measurement consistency compared to standing poses.

Accuracy Analysis Report

Strengths

- Fast and efficient landmark detection.
- High accuracy for front-facing full-body images.
- Consistent detection of major skeletal points.
- Suitable for real-time virtual try-on applications.

Limitations

- Sensitive to severe body rotations.
- Performance decreases when landmarks are occluded.
- Pixel-based measurements require scaling for real-world accuracy.
- Loose clothing can affect landmark visibility.

Overall Accuracy Assessment

Pose Type	Landmark Accuracy
Standing Front Pose	Very High
Arms Extended	High
Bent Arm Pose	High
Side Pose	Moderate
Walking Pose	Moderate

Measurement Improvement Report

The study shows that incorporating body measurements extracted from landmarks can significantly improve virtual try-on realism.

Benefits of Measurement-Based Fitting

- Improved garment scaling according to body dimensions.
- Better alignment of clothing around shoulders and waist.
- Reduced distortion during garment warping.
- More realistic garment placement on different body types.
- Enhanced personalization for individual users.

Recommended Improvements

- Combine pose detection with body segmentation for higher accuracy.
- Use multiple images for more reliable measurements.
- Apply depth estimation to improve real-world scaling.
- Integrate measurement calibration techniques for accurate sizing.

Conclusion

MediaPipe Pose successfully detected body landmarks and enabled estimation of important body measurements such as shoulder width, height, and waist position. The results indicate that body measurement extraction can enhance garment alignment, scaling, and realism in virtual try-on systems. Although certain pose variations affect measurement accuracy, the technology provides a strong foundation for building intelligent and personalized virtual fitting solutions.